Phone Stand Design: Problem Statement

Design a simple and functional phone stand that securely holds a standard smartphone in both portrait and landscape orientations. The stand must be compact, lightweight, and suitable for use on flat surfaces such as desks or tables. It should have a stable base, provide a comfortable viewing angle, and be easy to manufacture using materials like plastic, wood, or metal. The design should aim for minimal material waste and support cost-effective production, making it ideal for large-scale distribution to students or office workers.

Application of Software Tools

CAD Software (SolidWorks or equivalent)

The CAD tool will be used to:

- Develop a 3D model of the phone stand, clearly showing dimensions, angles, and material thickness.
- Produce technical drawings with front, side, and top views, including all necessary measurements for manufacturing.
- Allow visualization of the stand in both portrait and landscape orientations, and check the stability of the design.
- (Optional) Simulate basic stress or balance properties to ensure the stand will not tip over when holding a phone.

Deliverables:

- 3D model file
- Technical drawings with dimensions and labels

Microsoft Visio

Visio will be used to:

- Create a flow diagram of the manufacturing process. This might include steps such as material selection, cutting or printing, finishing (e.g. smoothing or polishing), quality control, packaging, and shipping.
- Create a user interaction flow diagram showing how a user unpacks, sets up, and uses the phone stand.

Deliverables:

- Manufacturing process flow diagram
- User flow diagram

MySQL Workbench (or equivalent database design tool)

Workbench will be used to:

- Design a database schema for managing design variants, production data, and inventory.
- Define tables that could include:
    - **Designs**: design_id, name, dimensions, material_type
    - **Production_Batches**: batch_id, design_id, material_type, quantity_produced, production_date
    - **Inventory**: item_id, batch_id, status (in stock, sold, defective), storage_location

Deliverables:

- ER diagram showing table relationships
- SQL code (optional) to create the schema